

Descriptive Statistics: Dispersion

Understanding How Data Spreads

MODULE 2 – PROBABILITY & STATISTICS

LESSON 2

B.TECH DATA SCIENCE & MACHINE LEARNING – 3RD SEMESTER

"Average tells us the **center**. Dispersion tells us how much the data **varies**. Knowing only the average is never enough."



The Opening Problem: Same Average, Different Behavior

Two ML models are evaluated on their prediction errors. On paper, they look identical. But are they really?

Model A — Prediction Errors

4, 5, 5, 5, 6

Mean = 5

Values cluster tightly around the mean. Predictions are **consistent and stable**.

Model B — Prediction Errors

1, 3, 5, 7, 9

Mean = 5

Values scatter widely. Predictions are **unpredictable and erratic**.

❓ Would you deploy Model A and Model B with equal confidence? Both have the same mean error — yet something is clearly different. What are we missing?

This is the gap that **dispersion** fills. The mean alone cannot tell the whole story.

What Is Dispersion?

Dispersion describes how far data values tend to spread away from the center. Two datasets can share the exact same mean yet behave completely differently.

Low Dispersion

Values are tightly packed near the center. Observations are similar to each other. The mean is a **reliable summary**.

High Dispersion

Values are widely scattered away from the center. Observations vary a lot. The mean alone is a **poor summary**.

Analogy: Imagine students standing around their teacher. Tightly packed students → low dispersion. Students scattered across the room → high dispersion. The **center** tells us WHERE. **Dispersion** tells us HOW SPREAD OUT.



Analogy: Two Classrooms, Same Average Score

Both classes scored an average of **70** on the mid-semester exam. Are they equally consistent?

Class A — Scores

68, 69, 70, 71, 72

Average = 70

Every student performed **close to the mean**. The class is consistent and predictable.

Class B — Scores

40, 55, 70, 85, 100

Average = 70

Performance ranges from failing to perfect. The class is **highly variable**.



The average score of 70 is **identical** in both classrooms. But a teacher, parent, or institution interpreting only the average would miss the critical difference in performance consistency. This is precisely why dispersion matters.

Range: The Simplest Measure of Spread

Definition

Range = Maximum - Minimum

Dataset: **10, 12, 15, 18, 20**

- Minimum = 10
- Maximum = 20
- Range = $20 - 10 = 10$

Analogy: Range measures the distance between the two outermost students standing in a line — it ignores everyone in between.

Properties of Range

✓ Easy to calculate

Requires only min and max values.

✓ Gives overall spread

Quickly shows the full extent of the data.

✗ Uses only 2 points

Ignores all intermediate observations.

✗ Sensitive to outliers

One extreme value completely changes it.

Why Range Can Be Misleading

A single extreme value can dominate the range, making two very different datasets appear similarly spread.

Dataset A

10, 11, 12, 13, 14

Range = $14 - 10 = 4$

Values are uniformly close together. Range accurately reflects the tight spread.

Dataset B

10, 11, 12, 13, 100

Range = $100 - 10 = 90$

One outlier (100) completely distorts the range. Four of five values still cluster near 10–13.

 Range only sees the extremes. It tells us nothing about where the bulk of the data actually lies.

We need a measure that uses **every observation**, not just the minimum and maximum. What if we measured how far *each* value is from the mean?

Variance: Measuring Distance From the Mean

Before the formula – understand the intuition. We want to measure how far each observation deviates from the mean.

The Problem with Raw Deviations

Dataset: **2, 4, 6, 8** → Mean (μ) = 5

x_i	$x_i - \mu$
2	-3
4	-1
6	+1
8	+3
Sum	0

Positive and negative deviations always cancel out – the sum is always 0.
We must **square** each deviation.

Population Variance Formula

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

x_i	μ
Each individual observation	Population mean
N	Σ
Total number of observations	Sum over all observations

Variance: Step-by-Step Calculation

Dataset: **2, 4, 6, 8** — Mean (μ) = 5. Let's compute variance one step at a time.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
2	$2 - 5 = -3$	$(-3)^2 = 9$
4	$4 - 5 = -1$	$(-1)^2 = 1$
6	$6 - 5 = +1$	$(+1)^2 = 1$
8	$8 - 5 = +3$	$(+3)^2 = 9$
Σ	0	20

Final Calculation

$$\sigma^2 = \frac{20}{4} = 5$$

Variance = 5

What Does "5" Mean?

On average, each observation's **squared distance** from the mean is 5. A higher variance means data is more spread out. A variance of 0 would mean every value equals the mean — no spread at all.

Why Do We Square the Deviations?

This is one of the most important conceptual questions in descriptive statistics. There are three compelling reasons.

1

Prevent Cancellation

Without squaring, positive and negative deviations always sum to zero – giving no useful information about spread.

2

Penalize Large Deviations

Squaring amplifies large deviations much more than small ones. A deviation of 5 contributes 25 – five times more than a deviation of 1 (which contributes 1).

3

Mathematical Utility

Squared deviations connect variance to broader statistical theory, optimization, and ML loss functions.

Squaring in Action

Deviation	Squared
±1	1
±2	4
±5	25
±10	100

ML Connection

In machine learning, the **Mean Squared Error (MSE)** loss function uses this exact principle. Large prediction errors receive disproportionately large penalties, forcing the model to prioritize eliminating big mistakes over small ones.

Standard Deviation: Back to Original Units

Variance is mathematically powerful, but it lives in **squared units** — which are hard to interpret in practice.

The Problem with Variance

If your data is measured in ₹ (Rupees):

- Deviations are in ₹
- Squared deviations are in ₹²
- Variance is in ₹²

What does "₹² of spread" even mean? We need to return to the original unit of measurement.

The Solution

$$\sigma = \sqrt{\sigma^2}$$

Standard Deviation = Square root of Variance

Using our example: $\sigma = \sqrt{5} \approx 2.236$

Interpretation

With $\sigma \approx 2.236$ on our dataset {2, 4, 6, 8}:

A typical observation deviates from the mean by approximately **2.236 units** (in the original measurement scale).

Analogy

Variance is the mathematical engine working under the hood. **Standard deviation** is the human-readable dashboard display — same information, interpretable units.

Population vs. Sample: Two Different Contexts

Whether you are describing an entire population or estimating from a sample determines which formulas and symbols you use.

Population Parameters

Mean: μ (mu)

Variance: $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$

Std Dev: σ

Denominator: **N** (all observations)

Sample Statistics

Mean: $\bar{x}$ (x-bar)

Variance: $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

Std Dev: **s**

Denominator: **n - 1** (Bessel's Correction)

Why n - 1 in the Sample?

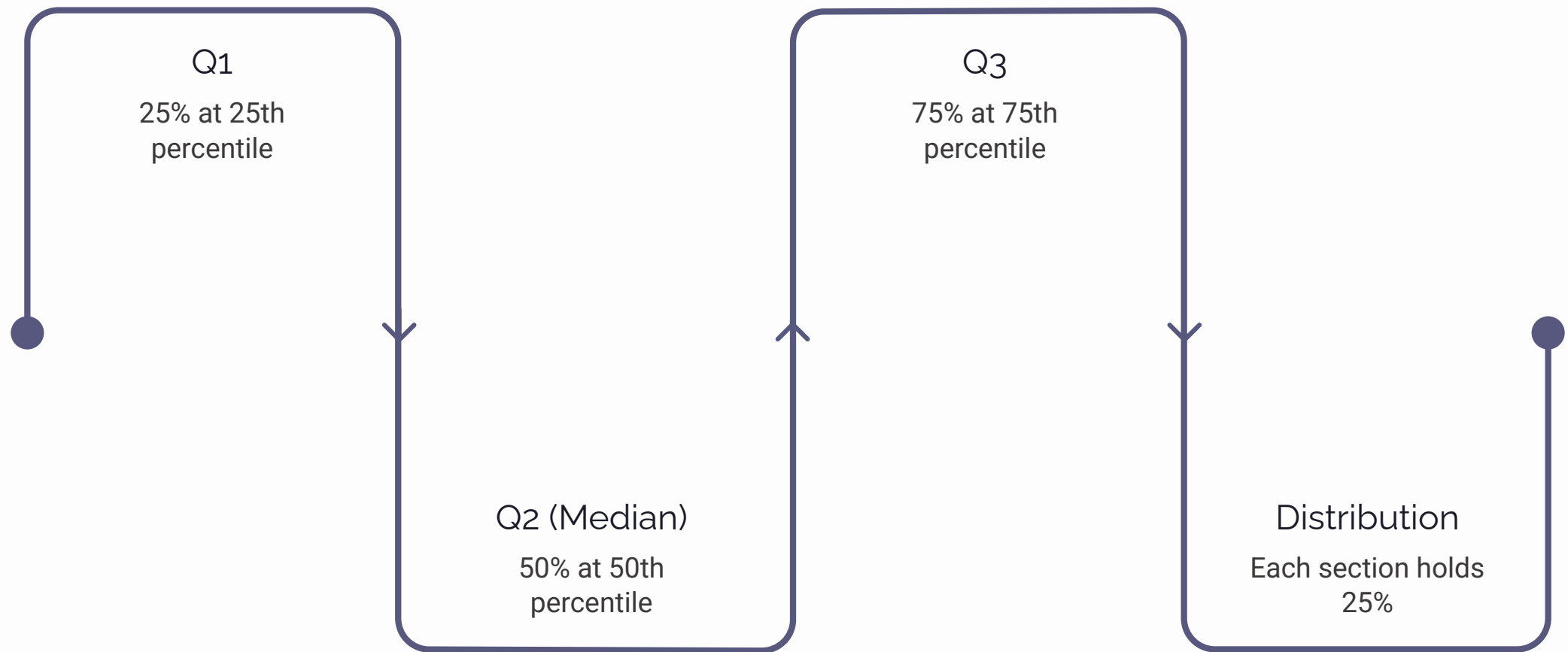
A sample tends to **underestimate** the true population spread — values near the center are more likely to be sampled. Dividing by **n - 1** instead of *n* corrects for this bias, producing an **unbiased estimator** of the population variance.

Practical Rule

In Data Science and ML, you almost always work with **samples** — a slice of a larger dataset or population. Always use **s²** and **s** when your data does not include the entire population of interest.

Quartiles: Dividing Data Into Four Equal Parts

While the mean and standard deviation describe center and spread globally, **quartiles** break the ordered data into four equal segments, revealing how the values are positioned.



Q1 — 25th Percentile

25% of all observations fall **at or below** Q1.
The lower quarter boundary.

Q2 — 50th Percentile

The **median**. Half the data lies below Q2, half above. The center of the ordered dataset.

Q3 — 75th Percentile

75% of all observations fall **at or below** Q3.
The upper quarter boundary.

i Always sort your data in ascending order before computing quartiles. Quartiles describe **position**, not distance — they tell you where values fall relative to the rest of the dataset.

IQR: Measuring the Middle 50%

Definition

$$IQR = Q3 - Q1$$

The **Interquartile Range** captures the spread of the central half of the data — it discards the bottom 25% and top 25%, focusing only on the middle 50%.

Example Calculation

- $Q1 = 10$
- $Q3 = 18$
- $IQR = 18 - 10 = 8$

The middle 50% of observations span a width of **8 units**.

Why IQR Is Powerful

- | | |
|--|---|
| → Outlier-Resistant | → Robust Summary |
| Extreme values on either tail do not affect the IQR at all. | Ideal for skewed distributions where mean and SD can mislead. |
| → Pairs with Median | |
| Just as SD complements the mean, IQR complements the median. | |

Analogy: Ignore the most extreme 25% on each side — study only the students standing in the middle of the group.



Box Plot: Seeing Dispersion at a Glance

The **box plot** (also called a box-and-whisker plot) encodes five key statistics in a single visual — making it one of the most efficient exploratory tools in data analysis.

The Box

Spans from **Q1 to Q3**. Represents the middle 50% of the data — the IQR.

The Line Inside

Marks the **median (Q2)**. Its position within the box reveals whether data is symmetric or skewed.

The Whiskers

Extend to the **most extreme non-outlier values** within $1.5 \times \text{IQR}$ of Q1 and Q3.

Individual Points

Points plotted beyond the whiskers are **potential outliers** — flagged for further investigation.

Outlier Detection Using the $1.5 \times \text{IQR}$ Rule

The most widely used rule for flagging potential outliers is based on the IQR. Values that fall beyond the **fences** are considered unusual.

The Fences

$$\text{Lower Fence} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Fence} = Q3 + 1.5 \times \text{IQR}$$

Worked Example

$Q1 = 10$, $Q3 = 18$, $\text{IQR} = 8$

Fence	Calculation	Value
Lower	$10 - 1.5(8)$	-2
Upper	$18 + 1.5(8)$	30

Any observation **below -2** or **above 30** is a potential outlier.

Critical Distinction

The $1.5 \times \text{IQR}$ rule **flags** observations — it does not make decisions. An observation beyond the fence is a **potential outlier**, not automatically an error.

Why 1.5?

The factor 1.5 was proposed by statistician **John Tukey**. For approximately normal data, this rule flags roughly 0.7% of observations — a practical balance between sensitivity and specificity.

Outlier ≠ Error: Domain Knowledge Is Essential

Statistics identifies **unusual observations**. What those observations *mean* is a question for domain expertise – not arithmetic.



₹50,000 Transaction Flagged

Could be **credit card fraud** – or a legitimate premium customer buying a laptop. Context determines the action.



Age = 150 in a Dataset

Almost certainly a **data entry error**. Humans do not live to 150. This warrants correction or removal.



Unusual Medical Measurement

Could indicate a **rare genuine condition**. Removing it could eliminate the most clinically important observation in the dataset.



ML Model Prediction Far From Normal

May signal a **model failure** on an unusual input – or a genuinely rare but valid real-world scenario.



Data Science mindset: "Flag first, investigate second, decide with domain knowledge third."

AI/ML Application: Model Error Dispersion

Let's return to the opening problem — now armed with the tools to analyze it rigorously.

Model A — Errors: 4, 5, 5, 5, 6

Mean = 5

Variance = $\frac{(1+0+0+0+1)}{5} = 0.4$

SD $\approx$ 0.63

Errors are tightly packed. Model A is **consistent and reliable**.

Model B — Errors: 1, 3, 5, 7, 9

Mean = 5

Variance = $\frac{(16+4+0+4+16)}{5} = 8$

SD $\approx$ 2.83

Errors are widely scattered. Model B is **unpredictable and unstable**.

Mean → Average Performance

How well does the model do on average?
Both models look identical here.

SD / Variance → Consistency

How stable is the model across different inputs? This is where Model A clearly wins.

IQR → Robustness

How does the model perform for the typical 50% of cases, ignoring extremes?

In ML model evaluation, **mean error alone is never sufficient**. A model that is accurate on average but wildly inconsistent may fail catastrophically in production.

Which Measure of Dispersion Should I Use?

Choosing the right measure depends on your data's characteristics and your analytical goal. Use this decision framework.

Measure	What It Tells Us	Outlier Sensitivity	Best Used When...
Range	Extreme-to-extreme extent	Very High	Quick first look; clean, outlier-free data
Variance	Avg squared deviation from mean	High	Mathematical modeling; loss functions in ML
Std Dev (σ or s)	Typical spread in original units	High	Interpreting spread; normally distributed data
IQR	Spread of central 50%	Low (Robust)	Skewed data; datasets with outliers

Symmetric / Normal Data

Use **Mean + Standard Deviation**

These two together fully characterize a normal distribution.

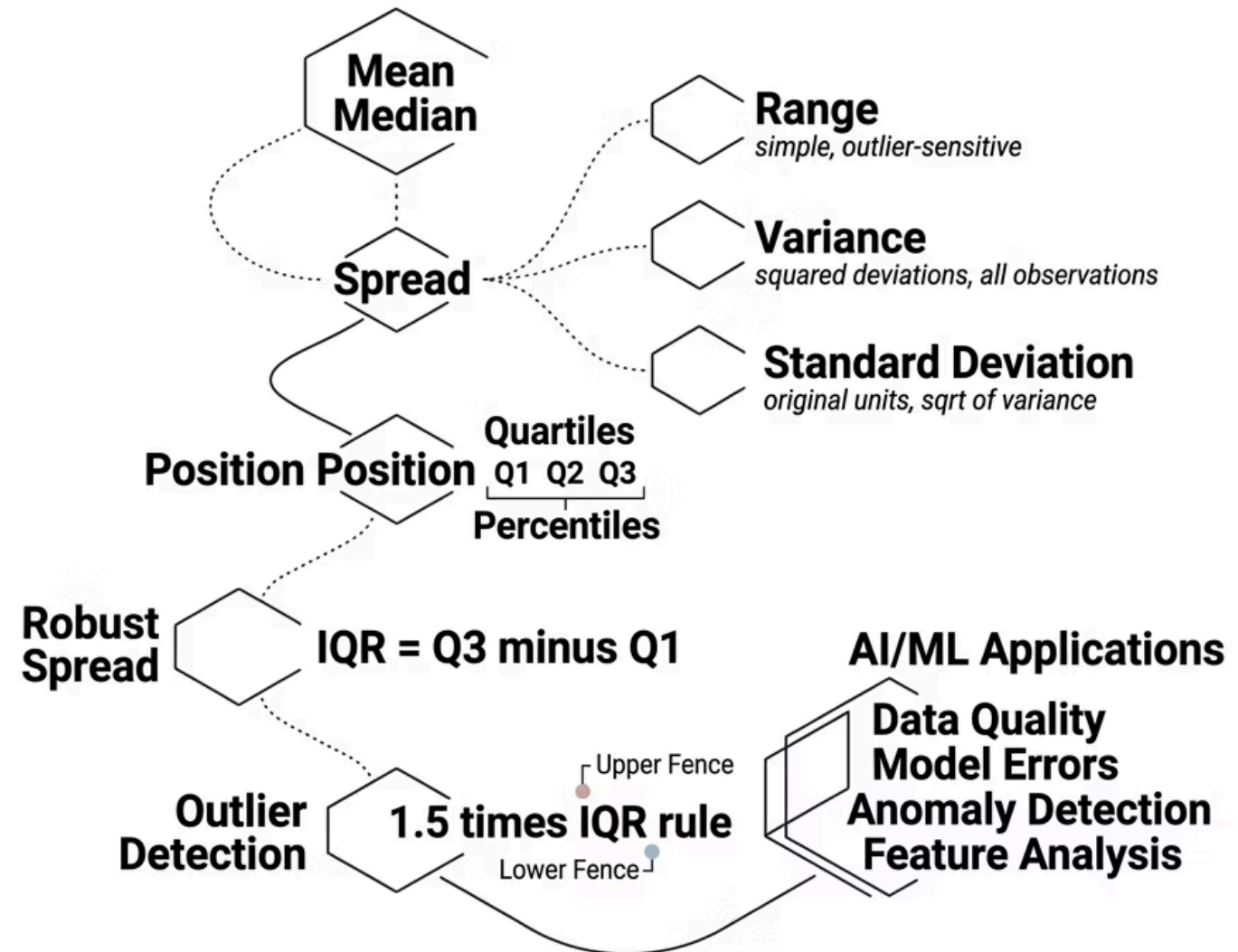
Skewed / Outlier-Prone Data

Use **Median + IQR**

These are resistant to extreme values and give a more honest picture of the typical observation.

Lesson Summary: The Complete Picture

Descriptive statistics requires both **center** and **spread** to tell the full story of a dataset.



- 1 Mean gives center; dispersion gives spread — you need both.
- 2 Range is simple but breaks down with even a single outlier.
- 3 Variance squares deviations so they don't cancel — but units are squared.
- 4 Standard deviation returns spread to the original, interpretable units.
- 5 IQR measures the middle 50% and is robust to extreme observations.

Think Like a Data Scientist

Both datasets below have a mean of exactly **50**. Use everything you've learned to analyze them.

Dataset A

48, 49, 50, 51, 52

Mean = 50

Dataset B

20, 35, 50, 65, 80

Mean = 50

1

Which has greater dispersion?

Compare the spread visually and think about standard deviation.

2

Which has the larger range?

Find max – min for both datasets.

3

Which has the larger standard deviation?

Compute or estimate σ for both.

4

If these were ML prediction errors, which model is more consistent?

Apply your understanding of model stability.

5

What visualization would you create first?

Histogram? Box plot? Scatter plot?

"Don't ask only: *What is the average?* Always ask: **How much does the data vary?** Knowing the center is the beginning – understanding the spread is where the insight lives."